APPLIED DATA SCIENCE(LAB)
ASSIGNMENT NO : 3

TITLE : Python for Data Handling: Data Visualization
NAME: Shinde Mahesh Tulshiram
ROLL NO : 150
CLASS : TY-B
GITHUB LINK :
https://github.com/maheshshinde9100/SCOE_CSE_SEM_6/blob/master/AppliedDataScienc e/Assignment3.py
DATASET : Global YouTube Statistics
https://www.kaggle.com/datasets/nelgiriyewithana/global-youtube-statistics-2023

Code:

```
# ============================================================
# APPLIED DATA SCIENCE LAB
# Assignment 3 – Data Visualization
# Dataset: Global YouTube Statistics 2023
# Student Name :  Mahesh Shinde
# ============================================================

# 1. Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# ----------------------------------------------------------
# 2. Load Dataset)
# ----------------------------------------------------------
df = pd.read_csv(
    "./datasets/Global YouTube Statistics.csv",
    encoding='latin1'
)

print("Dataset Loaded Successfully")
print("="*100)
# ----------------------------------------------------------
# 3. Clean Column Names
# ----------------------------------------------------------
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

print("Columns after cleaning:")
print(df.columns)
print("="*100)

# ----------------------------------------------------------
```

```
# 4. Basic Exploration
# --------------------------------------------------------
print("First 5 Rows:")
print(df.head())

print("\nDataset Shape:", df.shape)

print("\nMissing Values:")
print(df.isnull().sum())
print("="*100)

# --------------------------------------------------------
# 5. Data Cleaning
# --------------------------------------------------------

# Convert important numeric columns properly
numeric_columns = [
    'subscribers', 'video_views',
    'highest_yearly_earnings',
    'video_views_for_the_last_30_days'
]

for col in numeric_columns:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Fill missing numerical values with mean
df[numeric_columns] = df[numeric_columns].fillna(df[numeric_columns].mean())

# Fill categorical missing values with mode
categorical_cols = df.select_dtypes(include='object').columns
for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

print("Missing Values Handled Successfully")
print("="*100)

# --------------------------------------------------------
# 6. HISTOGRAM – Subscribers Distribution
# --------------------------------------------------------
plt.figure(figsize=(6,4))
plt.hist(df['subscribers'], bins=30)
plt.title("Distribution of Subscribers")
plt.xlabel("Subscribers")
plt.ylabel("Frequency")
plt.show()

# --------------------------------------------------------
# 7. BOXPLOT – Highest Yearly Earnings
```

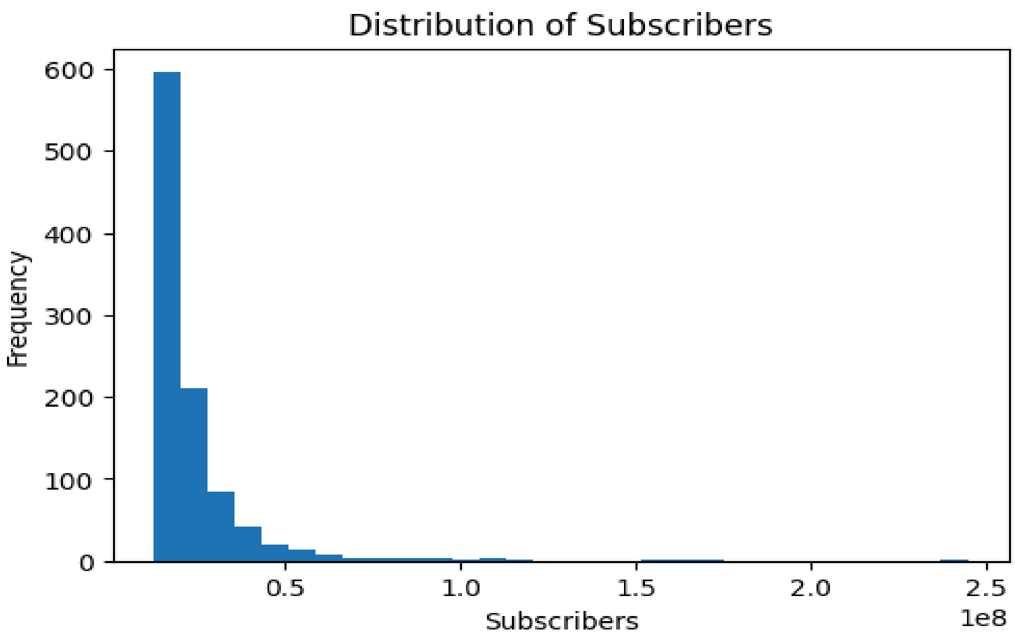
```
# --------------------------------------------------------
plt.figure(figsize=(6,4))
sns.boxplot(x=df['highest_yearly_earnings'])
plt.title("Boxplot of Highest Yearly Earnings")
plt.show()

# --------------------------------------------------------
# 8. SCATTER PLOT – Subscribers vs Video Views
# --------------------------------------------------------
plt.figure(figsize=(6,4))
plt.scatter(df['subscribers'], df['video_views'])
plt.title("Subscribers vs Video Views")
plt.xlabel("Subscribers")
plt.ylabel("Video Views")
plt.show()

# --------------------------------------------------------
# 9. BAR CHART – Top Channel Categories
# --------------------------------------------------------
plt.figure(figsize=(8,5))
df['category'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 YouTube Channel Categories")
plt.xlabel("Category")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()

print("Data Visualization Completed Successfully")
```

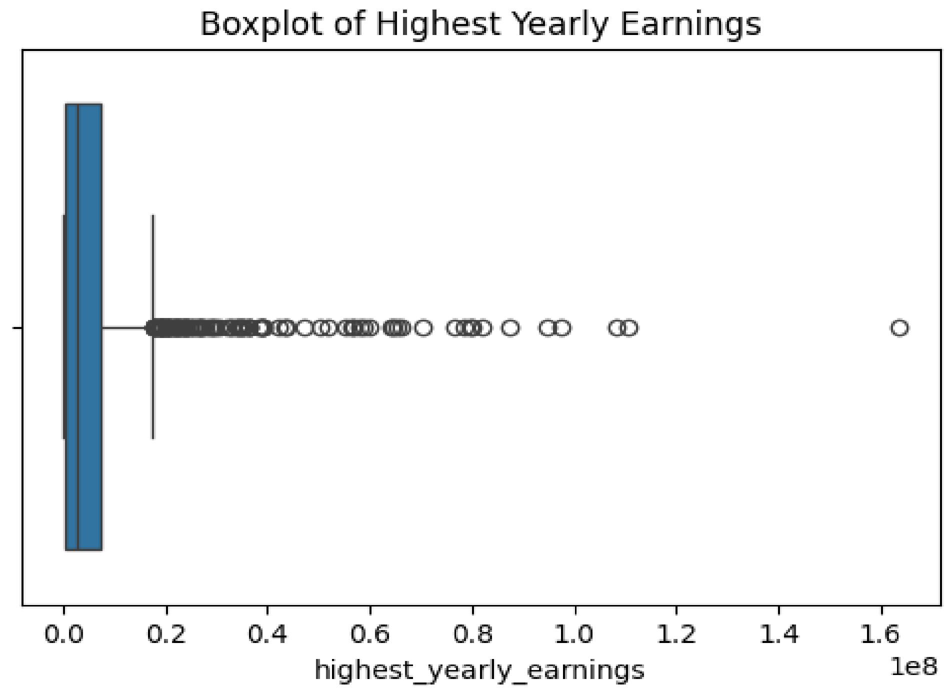
Output:

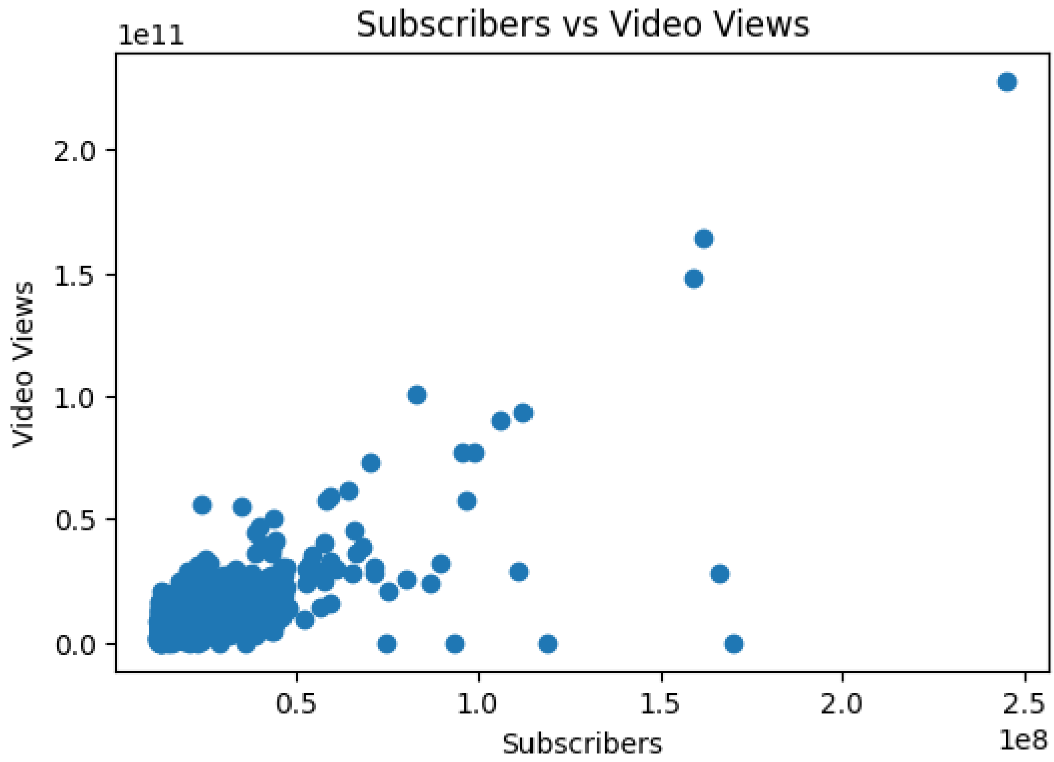
```
Dataset Loaded Successfully
======================================================================
Columns after cleaning:
Index(['rank', 'youtuber', 'subscribers', 'video_views', 'category', 'title',
       'uploads', 'country', 'abbreviation', 'channel_type',
       'video_views_rank', 'country_rank', 'channel_type_rank',
       'video_views_for_the_last_30_days', 'lowest_monthly_earnings',
       'highest_monthly_earnings', 'lowest_yearly_earnings',
       'highest_yearly_earnings', 'subscribers_for_last_30_days',
       'created_year', 'created_month', 'created_date',
       'gross_tertiary_education_enrollment_(%)', 'population',
       'unemployment_rate', 'urban_population', 'latitude', 'longitude'],
      dtype='object')
======================================================================
First 5 Rows:
   rank                youtuber  subscribers   video_views  \
0     1                T-Series    245000000  2.280000e+11
```

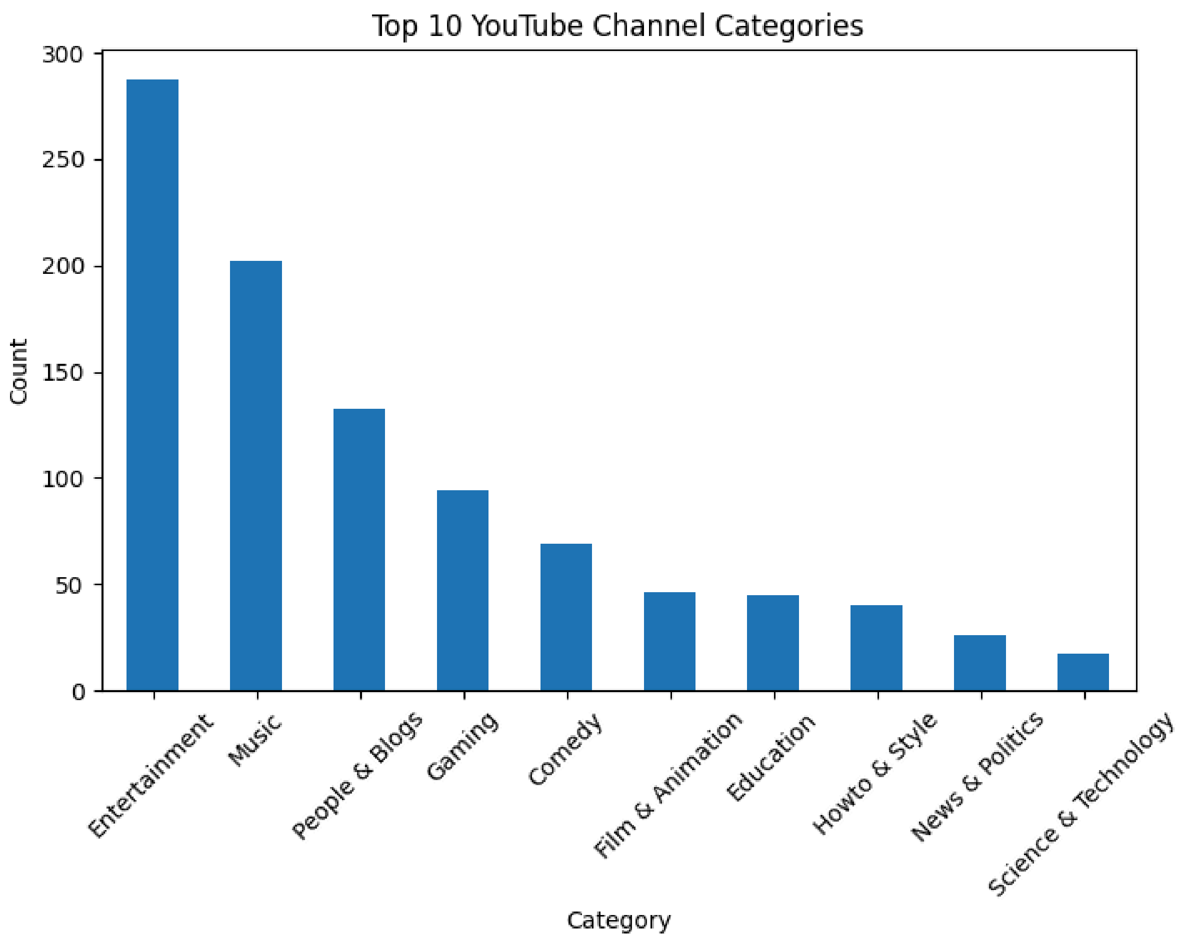
```
1     2           YouTube Movies    170000000  0.000000e+00
2     3                  MrBeast    166000000  2.836884e+10
3     4  Cocomelon - Nursery Rhymes    162000000  1.640000e+11
4     5                SET India    159000000  1.480000e+11

        category                   title  uploads       country  \
0          Music                T-Series    20082         India
1  Film & Animation            youtubemovies        1  United States
2    Entertainment                 MrBeast      741  United States
3       Education  Cocomelon - Nursery Rhymes      966  United States
4          Shows               SET India   116536         India

  abbreviation   channel_type  ...  subscribers_for_last_30_days  \
0           IN          Music  ...                     2000000.0
1           US          Games  ...                           NaN
2           US  Entertainment  ...                     8000000.0
3           US      Education  ...                     1000000.0
4           IN  Entertainment  ...                     1000000.0

  created_year  created_month  created_date  \
0       2006.0            Mar          13.0
1       2006.0            Mar           5.0
2       2012.0            Feb          20.0
3       2006.0            Sep           1.0
4       2006.0            Sep          20.0

  gross_tertiary_education_enrollment_(%)    population  unemployment_rate  \
0                                     28.1  1.366418e+09               5.36
1                                     88.2  3.282395e+08              14.70
2                                     88.2  3.282395e+08              14.70
3                                     88.2  3.282395e+08              14.70
4                                     28.1  1.366418e+09               5.36

  urban_population   latitude  longitude
0      471031528.0  20.593684  78.962880
1      270663028.0  37.090240 -95.712891
2      270663028.0  37.090240 -95.712891
3      270663028.0  37.090240 -95.712891
4      471031528.0  20.593684  78.962880

[5 rows x 28 columns]

Dataset Shape: (995, 28)

Missing Values:
rank                                        0
youtuber                                    0
subscribers                                 0
```

```
video_views                          0
category                        46
title                        0
uploads                          0
country                       122
abbreviation                       122
channel_type                         30
video_views_rank                         1
country_rank                      116
channel_type_rank                        33
video_views_for_the_last_30_days          56
lowest_monthly_earnings                      0
highest_monthly_earnings                     0
lowest_yearly_earnings                     0
highest_yearly_earnings                    0
subscribers_for_last_30_days             337
created_year                          5
created_month                           5
created_date                          5
gross_tertiary_education_enrollment_(%)    123
population                        123
unemployment_rate                        123
urban_population                      123
latitude                      123
longitude                       123
dtype: int64
============================================================================
Missing Values Handled Successfully
============================================================================
```




Boxplot of Highest Yearly Earnings
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1e8
highest_yearly_earnings


Subscribers vs Video Views
1e11
2.0
1.5
1.0
0.5
0.0
Video Views
0.5
1.0
1.5
2.0
2.5
1e8
Subscribers



Data Visualization Completed Successfully.